Project Report: Student Performance Analysis

1. Executive Statistical Summary

- **Mean Score**: 69.32 (Class average reflects an overall solid foundation).
- **Median Score**: 70.30 (Middle point shows consistent performance across the board).
- **Mode Score**: 70.30 (The most frequently occurring grade in the examination).
- **Standard Deviation**: 15.26 (Indicates a moderate spread of grades around the class average).
- **Variance**: 232.72 (Measures the variation and dispersion in student performance).
- **Score Distribution (Middle 50%)**: Between **58.30 (Q1)** and **79.90 (Q3)**.

2. Primary Performance Drivers (Correlation Analysis)

- **Study Hours vs Exam Score**: **(r = 0.43)** (Strongest correlation factor).
- **Attendance vs Exam Score**: **(r = 0.27)** (Weak to moderate positive impact).
- **Assignment Score vs Exam Score**: **(r = 0.11)** (Negligible linear relationship).
- *Insight*: Total dedicated **Study Hours** serves as the most critical factor influencing high exam outcomes.

3. Empirical Probabilities

- **Probability of Passing (Score greater or equal 50)**: **88%** (P = 0.88).
- **Probability of High Achievement (Score greater or equal 80)**: **24%** (P = 0.24).
- **Probability of High Dedication (Study more then 5 hours)**: **52%** (P = 0.52).

4. Outlier Analysis Findings

- **Z-Score Profile**: **STU001** is an anomaly with a critical negative Z-score of **-2.45**.
- **IQR Profile**: The standard mathematical Box Plot fence (1.5 * IQR) detected **0 outliers** (Lower Bound = 25.90).
- *Insight*: While technically inside the statistical boundaries, STU001 demands urgent attention because their exam performance drops drastically despite high attendance (95%) and study logs (9.5 hours).

5. Actionable Recommendations

- **Establish Mandatory Study Milestones**: Since study duration yields the highest returns (r=0.43), structured study groups should be arranged to help mid-tier students cross the 5-hour weekly threshold.
- **Conduct Student Wellness Interventions**: Target anomalous performance gaps (like STU001) early to identify underlying issues such as exam anxiety or digital submission technical errors.